

Dual Attention Multi-Instance Learning for Identifying Autism Spectrum Disorder Using Resting-State fMRI

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Abstract—Deep learning has been widely studied to develop computer-aided diagnosis methods for autism spectrum disorder (ASD) due to abnormalities in functional connectivity (FC) caused by neural activations. However, FC dysconnectivity occurs in limited regions of the brain, and the intricacy of large datasets for aggrandizing generalizability and variability is also responsible for an increase in heterogeneities. Thus, the crucial challenge for a classification framework is to identify and consolidate the distinctive features that can elevate the class separation. To resolve this predicament, we propose a dual attention multi-instance learning framework (DAMILF) for the diagnosis of ASD and its subtypes. The DAMILF comprises four modules: 1) identification of specific regions in functional magnetic resonance imaging (fMRI); 2) patch-level attention subnetwork for extracting and strengthening patch-based localized discriminatory features; 3) attention-aware feature map for balancing the corresponding share of each patch and generating a global feature space for the whole brain; and 4) subject-level subnetwork for supplementary whole brain global feature learning and ASD-related label prediction. Our proposed DAMILF was evaluated on large-scale baseline fMRI data collected from 408 ASD and 476 typical control subjects from the autism brain imaging data exchange (ABIDE-1) repository, and exhibited superior diagnosis performance and a better generalization ability compared with several state-of-the-art methodologies.

Impact Statement—ASD diagnosis is often delayed by years because symptoms and brain patterns vary widely, and objective

biomarkers are lacking. This diagnostic gap leads to missed early-intervention windows, prolonged family uncertainty, and healthcare inefficiency. We introduce DAMILF, an attention-driven artificial intelligence (AI) framework that derives interpretable whole-brain FC signatures from routine 5–10-minute resting-state fMRI scans. Unlike prior multimodal or handcrafted approaches, DAMILF learns diagnostically relevant connectivity patterns directly from data. Validated on the multisite ABIDE-1 cohort ($n = 884$), it achieves 86% accuracy in cross validation and maintains 81% average accuracy when tested on entirely held-out sites, demonstrating generalization across heterogeneous scanner types and acquisition protocols. The framework reveals age-related and subgroup patterns with neurobiologically interpretable attention maps. These generalizable biomarkers could complement behavioral assessments as objective screening tools, accelerate ASD subtype research, and enable more precise clinical-trial stratification.

Index Terms—ABIDE, attention mechanism, autism spectrum disorder (ASD), computer-aided diagnosis (CAD), functional magnetic resonance imaging (fMRI), multi-instance learning (MIL).

I. INTRODUCTION

Autism spectrum disorder (ASD) comprises hereditary neurodevelopmental disorders, including autism, Asperger's syndrome (AS), and related conditions [1], characterized by social communication deficits, restricted interests, and repetitive behaviors [2]. ASD prevalence is on the rise with an estimated 1.85% of American children experiencing ASD as reported by the Centers for Disease Control and Prevention [3]. Clinical diagnosis relies on *diagnostic and statistical manual of mental disorders* (DSMs) criteria [4], which are subjective and prone to misdiagnosis. Thus, the development of a computer-aided diagnosis (CAD)-based framework is crucial to determine the neurophysiology associated with ASD.

Resting-state functional magnetic resonance imaging (rs-fMRI) measures spontaneous blood-oxygen-level-dependent (BOLD) signal variations to capture functional connectivity (FC) patterns without task engagement [2]. Machine learning approaches have identified ASD-discriminative FC patterns [5], [6]. However, many of these studies are heavily dependent on the correct identification of atlas-based priori landmark locations (PLLs), exploiting the identical template space. Therefore,

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the performances of such studies are debatable in situations of wrongly identified PLLs [i.e., regions of interest (ROIs)]. Additionally, the PLL-based methods do not take into consideration all the prominent disease features scattered in the whole brain image, which may result in overall degraded diagnosis performance [7]. Similarly, the relationship between the PLL and the classification methodology requires extensive investigation, which is an additional obstacle. Since the distinguishable FC patterns between typical control (TC) and ASD groups vary in localized regions [8], only a limited number of regions have noticeable functional variations, while the rest exhibit diminutive distinctive details. Hence, the main task of the CAD-based methodology is to strengthen the contrastive features between TC and ASD groups by identifying discriminative short-range and long-range FCs in localized and comparatively prominent regions in the global image, respectively [9].

To overcome these problems, we propose a patch-based dual attention multi-instance learning framework (DAMILF) to locate prominent and discriminatory FC patterns for ASD diagnosis. The proposed DAMILF does not rely on any pre-defined ROI. Therefore, it is free from the defects caused by any potentially biased PLL-based methodology. The DAMILF comprises four stages, namely: 1) bag creation; 2) patch-level attention subnetworks (PASNs); 3) attention-aware feature map (AWFM); and 4) subject-level subnetwork (SSN). Each bag consists of eight image patches against a single rs-fMRI sample that belongs to either the TC or ASD group. The PASN facilitates DAMILF to select the localized discriminative FC patterns at the patch level. Then, the AWFM calculates and assigns an exclusive weight to each PASN feature. Moreover, the AWFM combines the patch-level features to construct a global feature space against a single rs-fMRI. Subsequently, the SSN exploits this global feature map and constructs a global subnetwork to diagnose ASD. Our proposed DAMILF classifier was trained and evaluated on the autism brain imaging data exchange (ABIDE-1) datasets for many ASD-related classification tasks, including ASD subtypes. Unlike existing works that use spatial domain or graph learning features, we employed chi-square maps from rs-fMRIs to train our DAMILF due to their ability to statistically assess the significance of FCs between different brain regions. The main contributions of this article are summarized as follows.

- 1) A data-driven masking technique is proposed to decisively stipulate the selection of prodigious FC patterns in an autistic brain.
- 2) A DAMILF combining PASNs for local feature discrimination and AWFM for adaptive patch weighting.
- 3) Comprehensive evaluation including generalizability analysis, age stratification, subtype identification, and ablation studies, demonstrating state-of-the-art performance.

Section II reviews related work; Section III describes materials and preprocessing; Sections IV and V present DAMILF and implementation; Section VI reports results and ablation studies; Section VII discusses findings and limitations; and Section VIII concludes. A comprehensive list of acronyms is provided in Table S-X of the supplementary materials.

II. RELATED WORK

Recently, there has been an increased focus on analyzing rs-fMRI to identify neurological disorders such as ASD and its subtypes through CAD-based frameworks. In this section, we succinctly summarize some of the previously reported conventional learning-based (CLB) and deep learning-based (DLB) studies to diagnose ASD patients. Then, we briefly describe the studies related to multi-instance learning (MIL) and the attention mechanism in the medical domain.

A. ASD Diagnosis With rs-fMRI

In a typical CAD-based framework, ASD is diagnosed by studying the brain's FC patterns in rs-fMRI. CLB studies exploit hand-crafted features to capture autistic FCs [6]. Kunda et al. [6] developed tangent Pearson embedding with domain adaptation for site-specific dependence, using ridge regression for classification. However, transductive learning limits generalization to unseen subjects. Wang et al. [10] used low-rank matrix decomposition with KNN/SVM for multisite adaptation.

Recently, many DLB studies have been developed for ASD diagnosis to resolve the challenges of generalizability and variability associated with CLB methods [5]. Ahmed et al. [11] proposed ensemble convolutional neural network (CNN) using glass brain features, though 4-dimensional (4-D)-to-2-D conversion could lose spatiotemporal information. Ji et al. [12] developed a model using graph-based hash learning ROIs as nodes and connections as edges, achieving 68.89% accuracy on ABIDE-1. Ji et al. [13] combined CNN with graphical lasso for sparse connectivity features. Both studies [12], [13] depend on accurate ROI identification.

B. MIL

MIL is a type of weakly supervised learning that comprehends training samples grouped in bags, while each bag comprises multiple instances with a single class label [14]. The MIL classifies the entire bag rather than the individual instances during a classification task. Thus, it is not possible to identify the individual instance labels inside the bag, but a positively labeled bag must have one positive instance at the very least. Positive bags may contain negative or ambiguous instances lacking class-specific characteristics. On the other hand, all negatively labeled bags must comprise negative instances. The MIL-based model does not necessitate manual annotations, which is a big advantage in any CAD framework.

The training process involves access to bag-level (i.e., global) labels to learn localized instance patterns, which are eventually utilized to classify bags with unknown labels. MIL has been applied to neuroimaging [15] and facial expression recognition [16]. Liu et al. [17] proposed landmark-based MIL for Alzheimer's diagnosis, concatenating local patch features. Similarly, Shao et al. [18] introduced transformer-based MIL for histopathology. The generation of bag-level global space from instance-level local features is a hitherto conundrum in MIL-based studies.

C. Attention Mechanism

Attention mechanisms identify and weight discriminative input subregions to enhance classification [19] through task-driven soft weights. Attention-based methods have improved medical imaging diagnosis and segmentation [19], [20]. Chen et al. [20] developed graph attention networks for ASD using weighted adjacency matrices. Despite their effectiveness, few studies combine MIL and attention. Adaptive instance weighting through attention mechanisms addresses the challenge of aggregating instance-level features into bag-level representations.

More recently, Zhu et al. [21] developed a dual attention multiple instance framework for diagnosing Alzheimer’s disease, which combined the MIL and attention mechanism. The authors used spatial attention blocks for discriminative feature extraction. An attention MIL pooling was explored to balance the relative contribution of these features and yield a global feature space. Subsequently, an attention-aware global module was proposed for classification. Similarly, Chikontwe et al. [22] studied an end-to-end dual attention multiinstance architecture for COVID-19 diagnosis. An attention-based pooling was explored for the selection of discriminative regions in the latent space. A spatial attention module in the spatial domain was exploited to learn the features from the selected regions. Later, a bag-level classification utilizing a contrastive loss was performed for diagnosis. Inspired by works [21] and [22], we propose a DAMILF for ASD diagnosis using rs-fMRI.

III. MATERIALS

This section introduces the demographic details of the studied rs-fMRI datasets and the adopted preprocessing pipeline in our work.

A. Studied Datasets

This study collected datasets from the open-access autism brain imaging data exchange (ABIDE-1) developed by the international neuroimaging data-sharing initiative [23]. ABIDE-1 is a collaborative baseline multisite consortium with data aggregated from 17 independent brain imaging laboratories worldwide. Despite mixed findings on connectivity range patterns, dysconnectivity in ASD is well established [24].

We employed 4-D spatiotemporal rs-fMRI data rather than precomputed 3-D maps or ROI-based approaches. Precomputed maps cannot measure voxelwise temporal synchrony in explicit neighborhoods, while ROI-based methods are vulnerable to misidentification and inadequate for whole-brain analysis. The 4-D time series (concatenated 3-D volumes) enables the generation of customized 3-D statistical maps quantifying synchronization between voxels and neighbors within radius R , capturing localized connectivity patterns for dysconnectivity assessment.

We employed the baseline whole brain ABIDE-1 4-D time series rs-fMRI datasets. This study included only those functional images that exhibited mean framewise displacement (MFD) less than 0.2 mm. Images with an MFD greater than

TABLE I
DEMOGRAPHIC INFORMATION OF SITE-SPECIFIC 884 STUDIED SUBJECTS FROM ABIDE-1 CONSORTIUM

Dataset Site	Gender (F/M)	Category TC/ASD	DSM_IV_TR AD/AS/PDD-NOS/ES	Recurrent Volumes
CALTECH	08/29	18/19	13/0/0/6	146
CMU	01/04	02/03	3/0/0/0	236–316
KKI	10/29	27/12	6/6/0/0	124–152
LEUVEN	07/54	34/27	27/0/0/0	246
MAX_MUN	04/38	24/18	1/17/0/0	116–196
NYU	35/136	98/73	51/18/4/0	176
OHSU	00/23	11/12	0/0/0/12	78
OLIN	05/20	11/14	0/0/0/14	206
PITT	07/38	23/22	22/0/0/0	196
SBL	00/26	12/14	2/6/6/0	196
SDSU	06/27	21/12	1/7/2/2	176
STANFORD	08/28	19/17	0/0/0/17	176–236
TRINITY	00/44	23/21	9/6/6/0	146
UCLA	08/67	39/36	36/0/0/0	116
UM	25/88	65/48	38/7/1/2	296
USM	00/61	23/38	37/0/1/0	232–236
YALE	14/34	26/22	5/5/12/0	196
TOTAL	138/746	476/408	251/72/32/53	N.A.

Note: Overall cohort: $N = 884$ (476 TC, 408 ASD); mean age: 17.21 ± 8.13 years; sex: 138 female (15.6%), 746 male (84.4%); ASD subtypes: 251 autistic disorder (AD, 61.5%), 72 Asperger’s syndrome (17.6%), 32 pervasive developmental disorder not otherwise specified (PDD-NOS, 7.8%), and 53 equivocal subtype (ES, 13.0%). Data were collected across 17 independent sites using heterogeneous scanner types and acquisition protocols. Detailed site-specific age distributions and full-scale intelligence quotient (FIQ) data are provided in Table S-I. Bold indicates the overall statistics computed across the complete dataset, including all sites.

0.2 mm are not suitable for a neural connectivity study due to the high values of motion artifacts, which corrupt the entire time series [25]. As a result, a total of 884 images that satisfied the MFD criteria were included in this study. The filtered cohort (476 TC, 408 ASD; ratio 1.17:1) maintains balanced class distribution, with the MFD criterion excluding approximately 20% of available ABIDE-1 baseline subjects without substantially altering TC/ASD proportions. Site-specific demographics are in Table I. Gender was recorded as female (F)/male (M), and subjects were classified as TC or ASD per *Diagnostic and Statistical Manual of Mental Disorders, 4th edition text revision* (DSM_IV_TR) criteria, with ASD subtypes: AD, AS, PDD-NOS, and ES for indecisive cases.

The number of 3-D recurrent volumes within a 4-D sample is also specified. The number of recurrent volumes across all dataset sites was not fixed, attributed to inter-site scanner variabilities. Moreover, several intra-site subjects also reported a variable number of recurrent volumes, probably ascribed to varied image capturing standards at different timepoints. The age and FIQ numbers are categorywise documented as mean $\pm$ standard deviation in Table S-I, while the ABIDE unique identifiers of the studied subjects are reported in Table S-II of the supplementary material.

B. Image Preprocessing

We used ABIDE-1 data preprocessed through the configurable pipeline for analysis of connectomes (CPACs), selected for superior classification performance [11]. This pipeline is composed of slice timing recalibration, motion correction, skull stripping, global mean-based intensity normalization to 10^4 ,

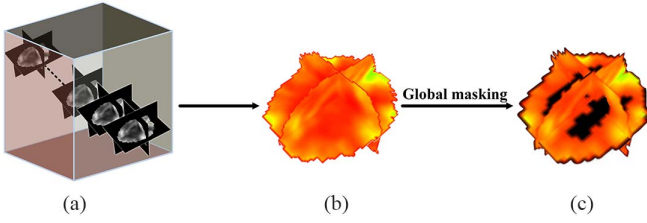


Fig. 1. Conversion of 4-D rs-fMRI time series imaging data to a 3-D masked statistical map with chi-square distribution. (a) 4-D rs-fMRI. (b) and (c) Unmasked and globally masked FCSMs with $R = 5$, respectively.

nuisance variable regression with 24 motion parameters, band-pass filtering (0.01–0.1 Hz), linear registration to anatomical space, and anatomical to standard Montreal Neurosciences Institute 152 template space that ensures spatial resolution of $3 \times 3 \times 3 \text{ mm}^3$, and a spatial Gaussian smoothing filtration with a 6 mm full width at half maximum.

Subsequently, every 4-D rs-fMRI time series was converted to a single 3-D statistical map by calculating Kendall's W (KW) per voxel in a customized neighborhood (R). We selected $R = 5$ for voxelwise neighborhood control based on empirical evaluation across multiple ($R \in \{1, 3, 5, 7, 9\}$; Supplementary Section S-IV; Fig. S-3), which demonstrated that $R = 5$ provides optimal balance between capturing local FC and maintaining robustness to noise. The voxel values of KW map lie in range $[0, 1]$, where 0 indicates the absence of any similarity trend while 1 means full agreement between recurrent volumes in time series. Furthermore, we calculated voxelwise Friedman chi-squared score from the 3-D KW map with the degree of freedom (DoF) equal to one less than the total number of recurrent volumes, which were present in the original 4-D time series. We exploited the *regional homogeneity (ReHo)* utility module developed by *Analysis of Functional NeuroImages (AFNI)* and powered by *Nipype* to generate KW and Friedman chi-squared maps (FCSMs) [26] as shown in Fig. 1.

FCSMs were selected for three reasons: 1) superior ASD diagnostic performance versus correlation-based matrices (ReHo: 73.57% versus CC200: 66.56% [27]); 2) standardized effect sizes facilitating cross-subject comparison despite heterogeneous scan durations (78–316 volumes); and 3) statistical significance quantification of temporal concordance. Ablation studies (Table VI) confirmed FCSM superiority over KW maps.

The Friedman transformation assumes nonparametric repeated measures with spatially independent continuous data. These are satisfied: rs-fMRI provides continuous within-subject measurements, the transformation handles arbitrary BOLD distributions, and $R = 5$ ensures local connectivity capture while maintaining independence across nonoverlapping regions.

The image resolution of the KW image and FCSM was recorded as $61 \times 73 \times 61$ voxels. Moreover, neurological disorders can be viewed as interruptions in voxelwise regular neural activity (VRNA). Inspired by the feature selection methodology by Huang et al. [5], we proposed a data-driven masking strategy (DMS) to identify the prominent and distinctive brain regions, exhibiting the most distinctive neural patterns linked with ASD.

The DMS comprises two stages: 1) identification of landmark locations; and 2) template dilation.

1) *Data-Driven Landmark Locations (DLLs)*: These are based on group comparisons and the required selection of distinctive brain regions, which can be tracked by location proposals (LPs). The LPs are based on two further conditions. We selected those voxels, which first demonstrated very large values of the VRNA compared to the corresponding average numbers in autistic brains. Second, the voxels exhibited group differences large enough to discriminate between the ASD and TC cohorts at the same time. We combined all ASD datasets to compute the mean voxel value in the FCSM, denoted as $\overline{\text{FCSM}}_{\text{ASD}}$. We also computed global mean and global standard deviation denoted as μ_{ASD} and σ_{ASD} for the ASD cohort. $\overline{\text{FCSM}}_{\text{ASD}}$ values higher or lower than the μ_{ASD} by an amount equal to σ_{ASD} were considered to fulfill the first selection criteria. Similarly, we exploited all TC datasets to compute $\overline{\text{FCSM}}_{\text{TC}}$, μ_{TC} , and σ_{TC} . Then, Hadamard product was calculated between the designated $\overline{\text{FCSM}}_{\text{ASD}}$ and $\overline{\text{FCSM}}_{\text{TC}}$ to combine both the conditions. This resulted in identifying the LPs. Eventually, the selected LPs which were prominent in the ASD group and obscured in the TC group constituted the template image (I_T). To identify ASD-specific landmark locations, we select voxels showing extreme values in ASD cohort but typical values in TC cohort, mathematically expressed by

$$I_T(i, j, k) = \left(\left(\left| \overline{\text{FCSM}}_{\text{ASD}}(i, j, k) - \mu_{\text{ASD}} \right| \geq \sigma_{\text{ASD}} \right) \odot \left(\left| \overline{\text{FCSM}}_{\text{TC}}(i, j, k) - \mu_{\text{TC}} \right| < \sigma_{\text{TC}} \right) \right) \quad (1)$$

where i, j , and k denote arbitrary voxel indices in the FCSM, while $|\cdot|$ and $\odot$ represent absolute values and Hadamard product, respectively. The I_T was thresholded by mapping all nonzero values to 1 and is mathematically described by

$$I'_T(i, j, k) = \begin{cases} 1, & \text{if } I_T > 0 \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

where the I'_T image represents the 3-D binary template and is shown in Fig. 2. The I'_T was recomputed separately for each cross-validation (CV) fold: all cohort means and standard deviations used in voxel selection were estimated from the fold's training subjects only, and the resulting I'_T fold-specific was applied to that fold's training and held-out test sets. The size of I'_T is $61 \times 73 \times 61$ voxels, and it contains a total of 1087 distinctive landmark locations.

2) *Template Dilation*: The KW hypothesis states that neurological activations are manifested by constellations instead of individual voxels. On the other hand, the generation of I'_T is based on voxelwise isolated neural activity. To this end, we applied the image dilation operation with a kernel size 5 to I'_T . This selects clusters of neighboring voxels around every LP and fulfills the KW criteria. Since neural activations manifest as clusters rather than isolated voxels, we apply binary dilation (kernel size 5) to select neighboring voxels around each landmark

$$I_{\text{GMT}}(i, j, k) = \bigvee_{i'=-2}^2 \bigvee_{j'=-2}^2 \bigvee_{k'=-2}^2 \text{KR}(i', j', k')$$

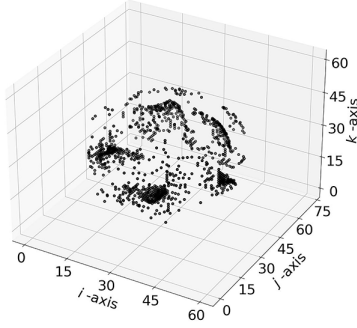


Fig. 2. Illustration of binary template image (I'_T) containing DLLs identified from the ASD and TC subjects in ABIDE-1. The I'_T consists of 1 and 0 s, where 1 s are represented with a black color and 0 s are made transparent to visualize DLLs in a 3-D spatial domain having dimensions of $61 \times 73 \times 61$. The example shown corresponds to an arbitrary fold-specific template recomputed from training data only.

$$\wedge I'_T(i + i', j + j', k + k') \quad (3)$$

where I_{GMT} image represents the 3-D globally masked template comprising DLLs that can act as reproducible biomarkers in a data-driven manner to identify the most distinctive brain regions associated with ASD. Further, we computed Hadamard product between the I_{GMT} and FCSMs to generate the globally masked dataset image (I_{GMD}), represented by

$$I_{GMD}(i, j, k) = I_{GMT}(i, j, k) \odot FCS(i, j, k). \quad (4)$$

Equation (4) captures voxelwise connectivity patterns from distinctive landmarks while suppressing heterogeneous information. The complete seven-step pipeline is detailed in Algorithm S-I (Section S-II in supplementary material).

IV. METHOD

The investigation of patch-level neural connectivity patterns is considered to be a multiinstance problem in this study. Therefore, we develop a CNN-based MIL framework for ASD diagnosis. A set containing N number of bags is created from the preprocessed training data, i.e., $\mathcal{D} = \{(I_{GMDn}, Y_n)\}_{n=1}^N$, where I_{GMDn} is the n th bag and Y_n is the corresponding bag-level (i.e., subject-level) label. Every bag comprises several unlabeled instances (i.e., patches), i.e., $I_{GMDn} = \{i_{n,m}\}_{m=1}^M$, where $i_{n,m}$ is the m th instance and M denotes the total number of instances in I_{GMDn} . In MIL, we label entire subjects (bags) but not individual patches (instances). A positive bag requires at least one positive instance; negative bag contains only negative instances and is defined by

$$Y_n = \begin{cases} 0, & \text{if } \sum_{m=1}^M l_{n,m} = 0 \\ 1, & \text{otherwise.} \end{cases} \quad (5)$$

The neural dysfunction occurs in certain localized brain regions, particularly in young children [5]. Thus, we consider the bag of patches to be positive if generated from an ASD patient. Similarly, a negative bag was created from a TC group image. As a result, the already available subject level labels were used as bag labels without involving any additional data processing operations.

While bag-level labels correspond to subject diagnoses (ASD/TC), our MIL formulation differs from standard supervised learning: 1) not all patches in ASD brains show discriminative dysconnectivity, but only localized regions [8], [24]; 2) patch-level labels are unknown and cannot be manually annotated due to heterogeneous dysfunction [5]; and 3) the dual attention mechanism must autonomously identify which patches carry diagnostic information. Positive bags (ASD) contain at least one discriminative patch among the eight selected, while negative bags (TC) contain only negative instances. This instance-level ambiguity within positive bags distinguishes MIL from supervised learning, enabling spatially localized pattern learning without patch-level annotations.

Our proposed DAMILF is shown in Fig. 3. The DAMILF consists of four sequential components, i.e., 1) bag creation; 2) PASNs; 3) AWFM; and 4) subject-level subnetwork. Briefly, a bag of patches is created and input to the PASNs, which yield instance-level features. These features are combined and processed to produce a bag-level attention-aware feature-map. Finally, this feature map is processed through an SSN to calculate a classification score for ASD diagnosis.

A. Framework Overview

The term “dual attention” in DAMILF refers to two complementary attention mechanisms that address heterogeneity at different scales. First-level attention operates within individual patches through spatial attention maps in the PASN module, identifying which specific FC patterns are most discriminative within localized brain regions. Second-level attention operates across patches through the AWFM module, which adaptively weights each patch’s contribution to the global feature representation based on learned discriminability scores and inter-patch relationships. This dual-level design enables DAMILF to capture both localized FC abnormalities and their relative importance for whole-brain ASD diagnosis.

Our approach differs from recent attention-based MIL frameworks for neuroimaging [21], [22] in three key aspects. First, we employ DLLs (Section III-B1) rather than predefined anatomical atlases, eliminating atlas dependency and improving generalizability across heterogeneous multisite data. Second, our AWFM integrates patch-level scores with dual global pooling operations to dynamically balance feature contributions, whereas prior methods use sequential attention or simple averaging. Third, we utilize FCSMs that statistically quantify voxelwise synchronization patterns, providing more robust features for multisite fMRI analysis than raw BOLD signals or Pearson correlation matrices.

B. Bag Creation

The selected DLLs are nonuniformly distributed in the brain image as shown in Fig. 2. It stipulates the need to select those brain regions that contain a higher number of DLLs to reduce heterogeneities. Motivated by patch-based studies [17], [21], we propose a method to extract k number of the most informative patches per subject. As not all patches contain an equal amount of discriminative information for ASD

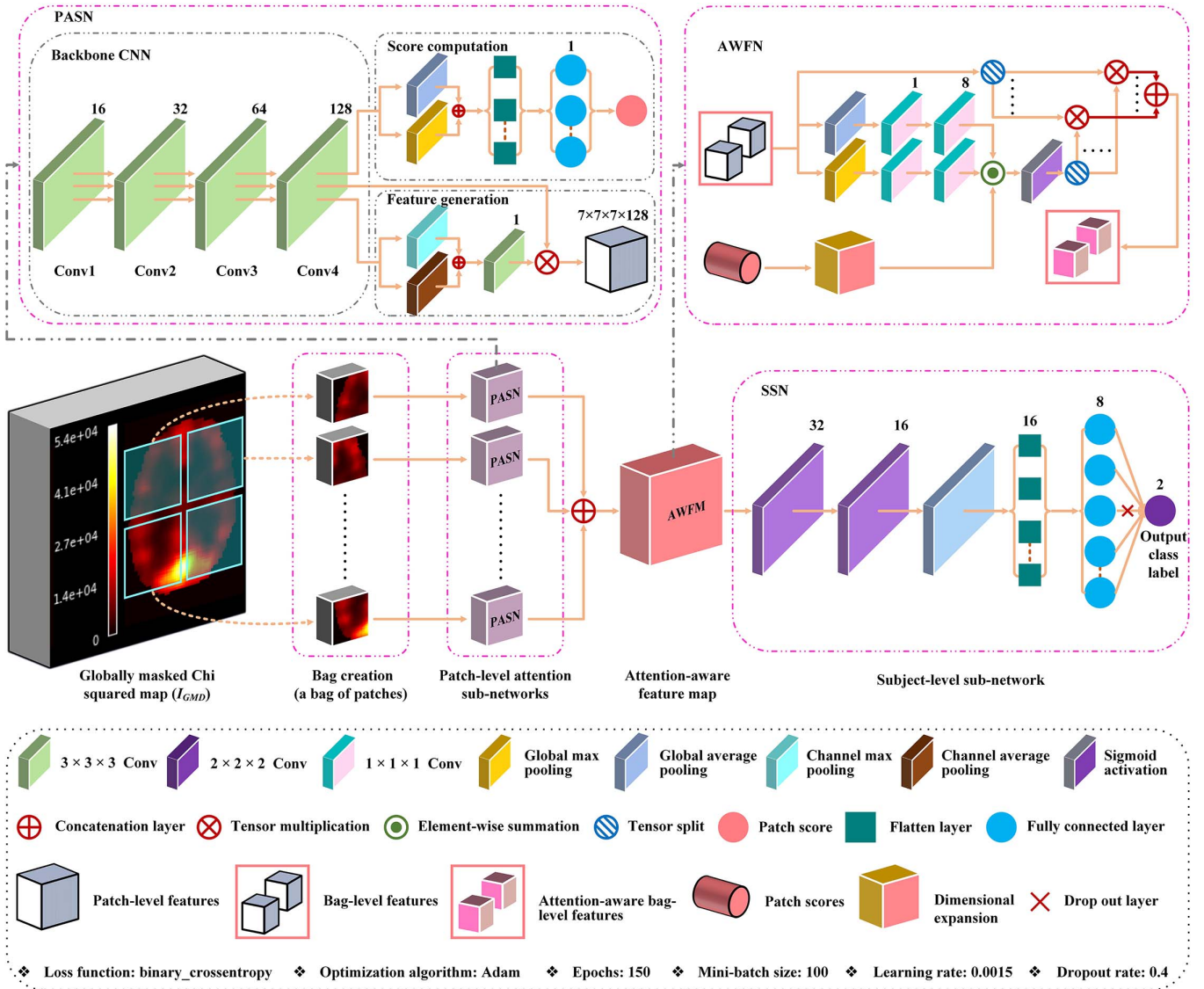


Fig. 3. Schematic diagram of the DAMILF for ASD diagnosis. The framework processes eight patches from DLL-rich regions through PASNs (spatial attention) and AWMF (adaptive patch weighting), generating a unified global representation for subject-level classification. This dual attention design addresses two challenges: only localized regions show ASD dysconnectivity, and patches contribute unequally to diagnosis. Color-coded legend indicates operations.

identification, we crop the input FCSMs to $60 \times 60 \times 60$ voxels. This provides a couple of advantages. First, the removed regions comparatively exhibit lower landmark density as they contain only 39 of 1087 landmarks. Similarly, a lower density indicates less informative regions. Second, the cropped FCSMs provide all three spatial dimensions to be uniform. This helps in partitioning the maps uniformly into multiple nonoverlapping cubical patches, each being the same size (i.e., $W \times W \times W$ and $W = 15$). The number of DLLs per patch with the patch's selection status for this study is discussed in Section S-VI of the supplementary materials. In this study, we selected all those patches that contain the number of landmarks greater than 50. This threshold was chosen based on the landmark distribution across all 84 patches (Table S-III): the counts are highly right-skewed (mean: 12.94, median: 3.0, standard deviation ≈ 20.55 , 90th percentile: ≈ 37.4 , 95th percentile:

≈ 56.85), with no patches falling in the 41–50 range. The >50 threshold thus represents a conservative upper tail cutoff (≈ 95 th percentile), isolating patches with exceptionally high landmark density while maintaining sufficient sample size ($n = 8$) for MIL. Importantly, the neurobiological validation in Section VI-F confirms that these eight high-density patches correspond to established ASD-affected brain regions (bilateral temporal, frontal, insular, cerebellar), demonstrating that the threshold successfully identifies diagnostically relevant regions rather than arbitrary high-variance areas. In this way, a total of eight patches are selected (i.e., patch no. 6, 10, 26, 35, 36, 56, 57, and 73; $8/84 = 9.52\%$). Our proposed DAMILF method combines this set of eight patches (i.e., instances) to form a bag (i.e., $I_{GMD} = \{I_1, I_2, \dots, I_M\}$, where $I_m \in \mathbb{R}^{W \times W \times W}$ and M is the number of selected patches) as the input to the framework.

C. Patch-Level Attention Subnetworks

The structure of the PASN module is shown in Fig. 3. All the PASN modules developed in our DAMILF have an identical composition. The employment of PASNs serves several purposes, namely: 1) computing a patch score; and 2) learning patch-level features. The patch score measures the significance of a particular patch in predicting the overall bag label. This module also augments the selected discriminative features.

A PASN comprises three submodules. The first is the backbone-CNN that consists of four 3-D convolutional layers, where each layer has an identical kernel size of $3 \times 3 \times 3$. The number of channels for Conv1 to Conv4 are 16, 32, 64, and 128, respectively. We designed the backbone-CNN submodule to generate abstract features from a given patch. We set a unit stride without zero-padding, batch normalization, and rectified linear unit (ReLU) activations for every Conv layer. The output ($\mathbb{B}$) of backbone-CNN is fed to score computation and feature generation submodules, which can be represented as follows:

$$\mathbb{B} = \{\mathbb{B}_1, \mathbb{B}_2, \mathbb{B}_3, \dots, \mathbb{B}_C\} \quad (6)$$

where $\mathbb{B}_i \in \mathbb{R}^{s \times s \times s}$, $s \times s \times s$ is the size of the feature maps per channel, and C is the number of channels in Conv4.

The score computation submodule calculates the patch score (p) by employing global average pooling (GAP), global max pooling (GMP), flatten layer and a fully connected layer with the Tanh activation function. Zhu et al. [21] demonstrated the effectiveness of employing both GAP and GMP in parallel instead of a single pooling mechanism to develop more powerful discriminatory features. This submodule is used to localize distinctive FC locations and quantify the discriminative capability of every patch. The size of the patch score is $2 \times C$.

Inspired by the spatial attention block designed by Zhu et al. [21], we developed a feature generation submodule to study the involvement of a specific FC, as the contribution of the FCs from different pathological locations varies. First, we calculated max pooling and average pooling along the channel axis from $\mathbb{B}$, and concatenated the relevant outputs. The size of the concatenated output is $s \times s \times s \times 2$. The concatenated values are input to a CNN layer with a kernel size $3 \times 3 \times 3$ with one number of channels and yielded feature attention map $\mathbb{A}_{FC} \in \mathbb{R}^{s \times s \times s}$. We utilized sigmoid activation to limit the range from 0 to 1. The $\mathbb{A}_{FC}$ distinguishes those FCs that require more emphasis in comparison to those that need to be suppressed in a feature space. Later, we performed tensor multiplication between $\mathbb{B}$ and $\mathbb{A}_{FC}$ to generate a patch-level attention-aware feature vector ($\mathbb{F}$). The channel max pooling and channel average pooling are described in the following equations, respectively:

$$\mathbb{B}_{\max}^{l,h,w} = \max(\mathbb{B}_1^{l,h,w}, \mathbb{B}_2^{l,h,w}, \mathbb{B}_3^{l,h,w}, \dots, \mathbb{B}_C^{l,h,w}) \quad (7)$$

$$\mathbb{B}_{\text{avg}}^{l,h,w} = \frac{1}{C} \sum_{i=1}^C \mathbb{B}_i^{l,h,w}. \quad (8)$$

The concatenation of $\mathbb{B}_{\max}$ and $\mathbb{B}_{\text{avg}}$ yields $\mathbb{A}_{\text{patch}}$. To identify which FC locations require emphasis versus suppression,

we generate a spatial attention map, which is given as follows:

$$\mathbb{A}_{FC} = \varsigma(\mathbb{C}(\mathbb{B}_{\max} \oplus \mathbb{B}_{\text{avg}})) \quad (9)$$

where ς is the sigmoid activation function, $\mathbb{C}$ is the weight of the convolutional layer, and $\oplus$ is the concatenation operator. $\mathbb{F}$ can be represented as follows:

$$\mathbb{F} = \mathbb{B} \otimes \mathbb{A}_{FC} \quad (10)$$

where $\otimes$ is a tensor multiplication and $\mathbb{F} \in \mathbb{R}^{s \times s \times s \times c}$.

The PASN architectural design was guided by established best practices in 3-D medical image analysis and empirical hyperparameter validation. The choice of four convolutional layers (Conv1–Conv4) balances feature abstraction capacity with overfitting risk: shallower architectures (two to three layers) provide insufficient hierarchical feature learning for complex fMRI patterns, while deeper networks risk overfitting given our sample size ($n = 884$) and introduce computational overhead. The $3 \times 3 \times 3$ kernel size is standard for 3-D medical imaging [21], [28], providing local receptive fields suitable for capturing voxelwise FC patterns without excessive parameter counts. Larger kernels ($5 \times 5 \times 5$ or $7 \times 7 \times 7$) would increase computational cost substantially, while potentially over-smoothing fine-grained connectivity details. The progressive channel expansion scheme $\{16 \rightarrow 32 \rightarrow 64 \rightarrow 128\}$ was empirically validated through hyperparameter ablation (Supplementary Section S-III, Fig. S-2), testing four configurations: $\{8, 16, 32, 64\}$, $\{16, 32, 64, 128\}$, $\{32, 64, 128, 256\}$, and $\{64, 128, 256, 512\}$. Results showed minimal performance variation (ACC range: 0.8476–0.8556), confirming architectural robustness, with $\{16, 32, 64, 128\}$ achieving the best ACC and AUC results.

D. AWFM

The architecture of the AWFM is illustrated in Fig. 3. It is developed to quantify the contribution of a patch in decision making. We concatenate every $\mathbb{F}$ generated by a PASN to develop bag-level features ($\mathbb{F}_b$) and is given in

$$\mathbb{F}_b = (\mathbb{F}_1 \oplus \mathbb{F}_2 \oplus \dots \oplus \mathbb{F}_n) \quad (11)$$

where n denotes the total number of patches. The $\mathbb{F}_b$ is fed to GAP and GMP simultaneously. Subsequently, the outputs of these pooling layers are processed through two convolutional layers with the number of channels equal to 1 and 8, respectively, and a kernel size of $1 \times 1 \times 1$ as described in the following equations, respectively:

$$\mathbb{A}_{\text{avg}} = \text{ReLU}(\mathbb{C}_1 \text{ReLU}(\mathbb{C}_0 \text{GAP}(\mathbb{F}_b))) \quad (12)$$

$$\mathbb{A}_{\text{max}} = \text{ReLU}(\mathbb{C}_1 \text{ReLU}(\mathbb{C}_0 \text{GMP}(\mathbb{F}_b))) \quad (13)$$

where $\mathbb{C}_0$ and $\mathbb{C}_1$ denote the weights of two convolutional layers. Both $\mathbb{A}_{\text{avg}}$ and $\mathbb{A}_{\text{max}}$ are subject-level attention maps based on inter-patch relationships. Similarly, the patch score (p), which is calculated in the PASN module, also provides a patch-level attention map based on intra-patch features. We concatenate p from all the patches to get the combined patch scores ($\mathbb{P}$), which is mathematically written as follows:

$$\mathbb{P} = (p_1 \oplus p_2 \oplus \dots \oplus p_c). \quad (14)$$

First, we expend the dimensions of $\mathbb{P}$ to match the size of the subject-level attention maps and compute the elementwise summation between all three attention maps. Then, a sigmoid activation is applied at sum to generate a patch-attention map ($\mathbb{A}_{\text{patch}}$). To determine each patch's relative contribution, we combine three attention signals (global average, global max, and patch scores) into a unified weighting

$$\mathbb{A}_{\text{patch}} = \varsigma (\mathbb{A}_{\text{avg}} + \mathbb{A}_{\text{max}} + \mathbb{P}). \quad (15)$$

Then, the $\mathbb{A}_{\text{patch}}$ and bag-level features ($\mathbb{F}_b$) are split into n parts as $\mathbb{A}_{\text{patch}} = (\mathbb{A}_{\text{patch}_1}, \mathbb{A}_{\text{patch}_2}, \dots, \mathbb{A}_{\text{patch}_n})$ and $\mathbb{F}_b = (\mathbb{F}_1, \mathbb{F}_2, \dots, \mathbb{F}_n)$. A tensor multiplication is performed between each patch-level attention-aware feature vector ($\mathbb{F}_i$) and its corresponding patch-attention map ($\mathbb{A}_{\text{patch}_i}$). Subsequently, the product results are concatenated to generate attention-aware bag-level features ($\mathbb{F}_{b'}$)

$$\begin{aligned} \mathbb{F}_{b'} = & ((\mathbb{F}_1 \otimes \mathbb{A}_{\text{patch}_1}) \oplus (\mathbb{F}_2 \otimes \mathbb{A}_{\text{patch}_2}) \oplus \dots \\ & \oplus (\mathbb{F}_n \otimes \mathbb{A}_{\text{patch}_n})). \end{aligned} \quad (16)$$

The generated $\mathbb{F}_{b'}$ features neither rely on the most distinctive patches nor do they combine all the patches with equal weights. The attention mechanism adaptively weights patches based on discriminative FC patterns, reducing heterogeneities.

E. Subject-Level Subnetwork

The layer structure of SSN is shown in Fig. 3. The SSN module takes attention-aware bag-level features ($\mathbb{F}_{b'}$) as input and outputs the binary class label (Y). We employ two convolutional layers, which have 32 and 16 number of channels, respectively, to further identify interconnections between weighted inter-patch features. The kernel size for both layers is $2 \times 2 \times 2$ voxels. Batch normalization and ReLU activations are employed after each layer. Then, we utilize GAP to down-scale the feature space to $\mathbb{R}^{1 \times 1 \times 16}$. Further, a flattened layer, a dense layer with eight channels, a dropout layer with a dropout rate of 0.4 and an output dense layer with two channels are employed. The dropout rate was selected in terms of the best validation accuracy. Although batch normalization is extensively used to generalize our training process, a dropout layer is additionally exploited as a further safeguard against overfitting. We deploy the softmax activation function in the output layer. The SSN module learns the whole brain FC patterns through the attention-aware bag-level features for ASD diagnosis.

The relatively shallow SSN architecture (two convolutional layers) is intentionally designed for final feature integration rather than hierarchical feature extraction. Unlike the PASN, which must extract discriminative features from raw patch data through four convolutional layers, the SSN receives already-processed attention-weighted features ($\mathbb{F}_{b'} \in \mathbb{R}^{7 \times 7 \times 7 \times 128}$) that have been refined through multiple stages: 1) PASN backbone CNN (four conv layers); 2) spatial attention within patches (9)–(10); and 3) adaptive patch weighting [AWFM (15)–(16)]. At this stage, the features already encode both local discriminative connectivity patterns and global inter-patch relationships. The SSN's role is therefore to learn final subject-level integration

patterns rather than to extract new hierarchical features, requiring only modest additional capacity. A deeper SSN architecture would risk overfitting given the sample size ($n = 884$) and the high dimensionality of input features ($7 \times 7 \times 7 \times 128 = 6,272$ features per subject), potentially degrading generalization performance. The current configuration appropriately allocates most of the representational capacity to the PASN modules (four conv layers $\times$ eight patches) that perform the primary feature extraction task. The dropout layer provides additional regularization to prevent overfitting during subject-level classification.

V. IMPLEMENTATION

DAMILF was implemented in Keras on NVIDIA GeForce RTX 12 GB GPU, using Adam optimizer (learning rate = 1.5×10^{-3} , batch size = 100, max epochs = 150). Training time: ≈ 335 seconds per fold. Inference requires ≈ 27 ms per subject, with preprocessing time similarly excluded, enabling real-time clinical deployment. The time to perform the preprocessing operations was not included. During training, we evaluated the model on the held-out validation fold after each epoch and saved the checkpoint achieving highest accuracy on that fold, following standard k-fold CV practice. All hyperparameters were fixed a priori and remained constant across all folds. All experiments used random seed 7 for reproducibility. The total number of trainable parameters was 560 388. The 884 subjects were split into tenfolds, maintaining class balance and multisite representation to reduce inter-site variability (site-specific contributions: Table S-VIII). This folding strategy improves inter-fold homogeneity [27], [29]. For binary classification (ASD versus TC), we used binary cross-entropy loss ($\mathcal{L}$) with subject-level labels to update model weights (W) via backpropagation (17)

$$\begin{aligned} \mathcal{L}(W) = & -\frac{1}{N} \sum_1^N [(X_i \log(\mathcal{P}(Y_i|X_i; W))) \\ & + (1 - X_i) \log(1 - \mathcal{P}(Y_i|X_i; W))] \end{aligned} \quad (17)$$

where N is samples per fold and $\mathcal{P}(Y_i|X_i; W)$ is prediction probability. Loss backpropagates from SSN through AWFM to PASN modules.

Training convergence was monitored through validation accuracy across all folds, with early stopping implemented by retaining weights corresponding to peak validation performance. The model converged stably across all folds, typically within 70–130 epochs of the 150-epoch budget, indicating no training instability or premature stopping. Validation accuracy curves showed consistent monotonic improvement in early epochs followed by plateau, with no evidence of divergence. The combination of batch normalization (after each convolutional layer), dropout regularization, and early stopping provided adequate training stability.

VI. EXPERIMENTS AND ANALYSES

In this section, we first describe the experiment settings, illustrate the classification performance, and demonstrate the

generalization capability by comparing our model with several state-of-the-art studies. Then, we validate the efficacy of the predefined model components, including the use of the FCSM, the selection of neighborhood, DLLs, and template dilation. Moreover, we also measure the impact of the model parameters (e.g., the dimensions and number of patches in a bag). Finally, we evaluate our model performance to identify the possible ASD subtypes.

A. Experiment Settings

We validated the performance of our proposed model on a couple of ASD-related classification tasks, such as ASD diagnosis (i.e., ASD versus TC) and ASD subtype identification (i.e., AD versus AS versus PDD-NOS). We used accuracy (ACC), sensitivity (SEN), specificity (SPE), and area under the receiver operating characteristic (ROC) curve (AUC). These metrics are described as: $ACC = (T_P + T_N) / (T_P + F_P + T_N + F_N)$, $SEN = T_P / (T_P + F_N)$, and $SPE = T_N / (T_N + F_P)$, where T_P , T_N , F_P , and F_N symbolize true positive, true negative, false positive, and false negative, respectively. We computed AUC by identifying all possible pairs of the true positive rate (SEN) and the false positive rate ($1 - SPE$), calculated by varying the threshold over the prediction scores yielded by our DAMILF model. The datasets for the ASD subtypes are not balanced. Thus, we use $F1$ -measure and macro-average accuracy (MAC) as the evaluation metrics during subtype identification to globally measure the performance, which are defined as: $F1\text{-measure} = T_P / (T_P + (1/2)(F_N + F_P))$ and $MAC = (1/c) \sum_1^c (ACC)$, where c denotes the number of ASD subtypes.

B. Competing Methods

We compared DAMILF against five state-of-the-art methods evaluated on identical subjects ($n = 884$): 1) tangent pearson embedding to solidify maximum independence domain adaptation (TP-MIDA) [6]; 2) dual-stream multiple instance learning (DSMIL) network [30]; 3) deep multimodal neuroimaging framework (DeepMNF) in single modality mode [27]; 4) uncertainty-based Instance eXclusion (UBIX) MIL framework [31]; and 5) ensemble learning framework (ELF) [11]. These methods differ in input representation (atlas-based versus data-driven versus 2-D slices), attention mechanisms (none versus single-level versus dual), and multisite handling. Detailed methodological comparison is in Table S-VII reported in Section S-IX (supplementary material).

C. Diagnostic Performance

We evaluated the tenfold CV performance of our DAMILF method and competing techniques to ensure a fair comparison for ASD diagnosis (i.e., ASD versus TC), which is reported in Table II, where the best performing metrics are bolded. Our DAMILF method outperforms the competitors in all four studied metrics [i.e., $ACC = 0.8556$ (95% confidence interval (CI): 0.8398–0.8714), $SEN = 0.8553$ (95% CI: 0.8392–0.8714), SPE

TABLE II
TENFOLD CV COMPARATIVE RESULTS BETWEEN OUR DAMILF METHOD AND STATE-OF-THE-ART TECHNIQUES FOR ASD CLASSIFICATION (I.E., ASD VERSUS TC)

Method	Reference	ACC	SEN	SPE	AUC
TP-MIDA	Kunda et al. [6]	0.7234	0.7260	0.7218	0.7819
DSMIL	Li et al. [30]	0.7268	0.7283	0.7294	0.7629
DeepMNF	Abbas et al. [27]	0.7387	0.7415	0.7358	0.7670
UBIX	Vente et al. [31]	0.7612	0.7658	0.7684	0.7977
ELF	Ahmed et al. [11]	0.8121	0.8236	0.8193	0.8537
DAMILF	Our method	0.8556	0.8553	0.8560	0.9201

Note: Bold indicates the best-performing metric values among all compared methods.

$= 0.8560$ (95% CI: 0.8411–0.8718), and $AUC = 0.9201$]. The narrow CIs for ACC, SEN, and SPE (CI widths: 0.0316, 0.0322, and 0.0307, respectively) confirm low inter-fold variability and high reproducibility of the proposed framework.

To assess robustness to random initialization, we repeated the tenfold CV with two additional random seeds (42, 123). The performance across all three seeds (7, 42, 123) was: $ACC = 0.8556 \pm 0.0018$, $SEN = 0.8553 \pm 0.0021$, $SPE = 0.8560 \pm 0.0016$, $AUC = 0.9201 \pm 0.0024$. The low standard deviations confirm result stability across different random initializations.

Additionally, we studied the ROC trends for each fold of the tenfold CV performance for DAMILF, which is shown in Fig. 4. We calculated the 95% CIs for each fold by first computing the AUC from the false positive rate and true positive rate of the predictions, then estimating standard error using sample counts and variance estimates; and finally determining confidence bounds using the z-score of 1.96. Fold-specific CIs range from [0.7857, 0.9453] for Fold 04 (worst case) to [0.9012, 0.9992] for Fold 01 (best case), demonstrating consistent AUC performance across independent validation sets. The mean ROC curve with AUC value of 0.9201 is illustrated with a dark blue curve, while the grey area represents one standard deviation.

D. Leave-One-Site-Out Cross-Validation (LOSOCV) Performance

To demonstrate the robustness, generalizability, and applicability of our DAMILF method to new unseen data, we performed LOSOCV. We implemented LOSOCV by employing data from one site as an independent validation set, while the data from all other sites were treated as a training set. The LOSOCV performances for all 17 sites alongside the global mean values are presented in Table III. The values of global mean ACC, SEN, SPE, and AUC are 0.8115, 0.8183, 0.8047, and 0.8193, respectively. We identified three sites, namely CMU, KKI, and OHSU, which exhibited exceptional performance. The studied metrics for these sites lie above one sigma. The CMU site with 100% values is an outlier because it contains only five samples of the 884 studied images. Similarly, we also found a couple of sites with evaluation metrics lower than one sigma, i.e., UCLA and UM. These two sites indicate the absence of inter-site homogeneities and the presence of site-specific mutability. Largely, the results reported in Table III show the efficacy of the DAMILF model in overcoming the challenges of overfitting and alleviating the inter-site heterogeneities.

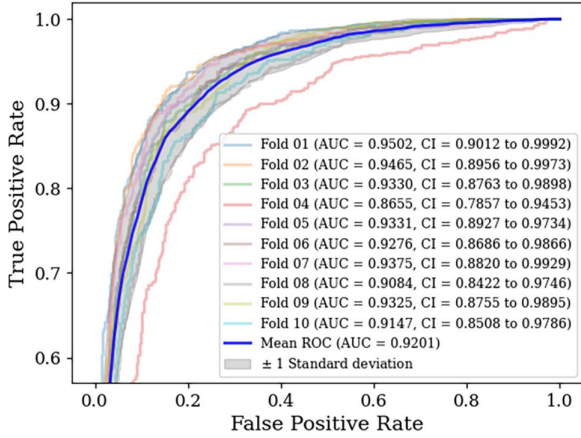


Fig. 4. ROC curves demonstrate consistent diagnostic performance across tenfold CV with narrow 95% CIs. The tight clustering around the mean (dark blue) with small standard deviation (gray) indicates stable generalization across different splits rather than overfitting to specific data subsets.

TABLE III
LOSOCV PERFORMANCE FOR ALL 17 INDEPENDENT
SITES FROM THE ABIDE-1 DATASETS

Site	ACC	SEN	SPE	AUC
CALTECH	0.8243	0.8108	0.8378	0.8218
CMU	1.0000	1.0000	1.0000	1.0000
KKI	0.9615	0.9744	0.9487	0.9887
LEUVEN	0.7459	0.7377	0.7541	0.7666
MAX_MUN	0.7976	0.7857	0.8095	0.7827
NYU	0.7456	0.7602	0.7310	0.7887
OHSU	0.9130	0.9130	0.9130	0.9057
OLIN	0.8200	0.8400	0.8000	0.8352
PITT	0.7889	0.7778	0.8000	0.7644
SBL	0.7500	0.7692	0.7308	0.7517
SDSU	0.8939	0.9091	0.8788	0.8597
STANFORD	0.7639	0.8611	0.6667	0.7418
TRINITY	0.8068	0.8182	0.7955	0.8431
UCLA	0.7133	0.7067	0.7200	0.7289
UM	0.6947	0.6903	0.6991	0.7151
USM	0.7951	0.7869	0.8033	0.8403
YALE	0.7813	0.7708	0.7917	0.7937
Global mean	0.8115	0.8183	0.8047	0.8193

Note: Bold indicates the global mean statistics aggregated across all 17 independent sites from the studied datasets.

The substantial performance variation across sites (ACC range: 0.6947–1.0000) provides valuable insight into the framework’s robustness against scanner and protocol variability. Three sites achieved exceptional performance: KKI (ACC = 0.9615, $n = 39$), OHSU (ACC = 0.9130, $n = 23$), and CMU (ACC = 1.0000, $n = 5$). Notably, KKI’s robust performance with a moderate sample size suggests favorable data quality, while CMU’s perfect ACC should be interpreted cautiously, given the limited sample size. Conversely, two sites exhibited notably lower performance: UM (ACC = 0.6947, $n = 113$) and UCLA (ACC = 0.7133, $n = 75$). The performance variability does not correlate simply with sample size (e.g., NYU with $n = 171$ achieves ACC = 0.7456, while smaller OHSU with $n = 23$ achieves ACC = 0.9130), indicating that site-specific factors such as scanner characteristics, acquisition protocols, subject demographics, and data quality contribute more substantially than sample quantity alone. Despite this heterogeneity, the

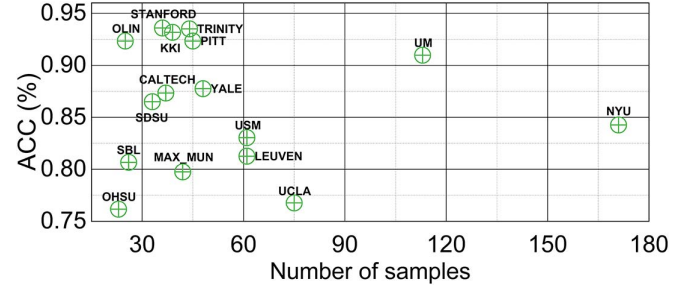


Fig. 5. Intra-site CV accuracies reveal that performance depends more on site-specific data quality than sample size. Despite varying sample sizes (23–171), performance does not correlate with quantity, OLIN ($n = 25$) achieves 0.9233 while larger NYU ($n = 171$) achieves 0.8426. This indicates scanner characteristics, acquisition protocols, and subject demographics impact performance more than sample quantity. Cross-referencing with LOSOCV (Table III) distinguishes distribution shift from internal heterogeneity.

global mean LOSOCV ACC (0.8115) remains within 0.0441 of tenfold CV performance (0.8556), demonstrating reasonable generalization capability across diverse scanner environments.

E. Intra-Site Tenfold CV Accuracy

We computed intra-site tenfold CV accuracies to study the site-specific variabilities. In this group of experiments, we exclusively selected all training and validation samples from a single site to perform tenfold CV and to study the site-specific homogeneities. We excluded the CMU site due to the small number of samples, i.e., five as reported in Table I, because we must have a minimum of ten samples from each site to get ten folds of data. A StratifiedKFold module from the Scikit-learn library [32] was utilized to randomly assign data samples to each fold while maintaining the original class balance of that site. This experiment was performed ten times, and the average accuracy per site was reported to alleviate the impact of biasedness due to random data sampling. The intra-site tenfold CV accuracy results are shown in Fig. 5, where the x-axis represents the number of samples, and the y-axis represents tenfold CV average accuracy for each site. Results do not follow the expected sample size-accuracy relationship. Scanner variability and diverse acquisition protocols within sites, alongside inadequate site-specific samples, prevented establishing clear size-accuracy correlations (Fig. 5).

These results are consistent with the literature as Kunda et al. [6] and Abbas et al. [27] also observed similar trends for intra-site CV performance. The STANFORD and OHSU sites exhibited the maximum and minimum accuracies with values of 0.9358 and 0.7617, respectively.

F. Attention Interpretability and Neurobiological Correspondence

To examine the neurobiological basis of learned attention allocation, we mapped the eight selected patches to the automated anatomical labeling 3 (AAL3) atlas [33] and analyzed cross-fold consistency. These patches corresponded to 20 unique AAL regions (Fig. 6), grouped by anatomical category as follows: bilateral temporal cortex (37.5% of aggregate attention

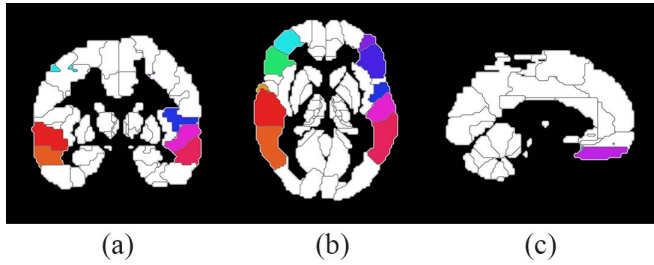


Fig. 6. DAMILF-identified patches map to 20 anatomically distinct AAL3 regions spanning temporal, frontal, insular, and cerebellar areas, shown in three orthogonal views (coronal, axial, sagittal). This spatial distribution aligns with established ASD neuroimaging literature reporting dysconnectivity in these regions, validating that the data-driven approach discovers clinically meaningful patterns rather than spurious correlations. The multiregional involvement supports the distributed network disruption theory of ASD pathophysiology. Refer to Supplementary Table S-IV for complete patch-to-region mapping. (a) Coronal view. (b) Axial view. (c) Sagittal view.

weight), frontal areas (25.9%), insula (25.0%), and cerebellum (11.7%). The spatial distribution of selected patches is shown in Fig. S-8; complete patch-to-region correspondence is detailed in Table S-IV.

Cross-fold stability analysis revealed excellent consistency of attention weights across all validation folds (Fig. 7), with CoV (standard deviation/mean $\times 100\%$) ranging from 8.4% (patch 36, frontal) to 15.1% (patch 35, insula), and mean CoV of $12.15 \pm 1.99\%$. Lower CoV indicates more consistent attention allocation across independent data splits. Patch 10 received the highest mean attention weight (0.149 ± 0.018), while patch 6 received the lowest among selected patches (0.111 ± 0.014). Attention weights exhibited strong positive correlation with landmark density (Pearson $r = 0.997$, $p < 0.0001$; Fig. S-6), confirming that the attention mechanism appropriately prioritized regions with statistically significant group differences (ASD versus TC), consistent with the data-driven landmark selection methodology (Section III-B1).

G. Influence of Age on Diagnostic Performance

We also studied the influence of age on the diagnostic performance of the proposed DAMILF to comprehend the level of heterogeneities in three exclusive age cohorts, i.e., children, adolescents, and adults. Therefore, the studied datasets were divided into three categories in accordance with the subject's age. The demographic details of all the studied cohorts are reported in Table IV. The age of every subject is recorded at the time of MRI scanning, and every scan is assigned to only one of the cohorts. The maximum value in age set is exclusive, e.g., the scans of subjects aged less than ten years are grouped to form the children cohort. Moreover, we also report gender and category (i.e., diagnosis label) of every subject alongside the inter-cohort age variabilities in terms of mean $\pm$ standard deviation.

We ran this experiment ten times and reported the average values of the performance metrics. The tenfold CV performance comparison between the three age cohorts is illustrated in Fig. 8.

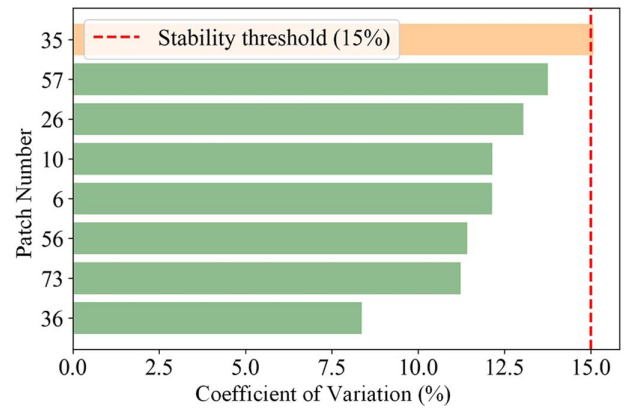


Fig. 7. Cross-fold stability of learned attention weights demonstrates consistent patch importance across tenfold CV. All eight patches exhibit coefficient of variation (CoV) below the 15% stability threshold (red dashed line). This low variability indicates that the dual attention mechanism consistently identifies the same diagnostically important regions across different data partitions rather than overfitting to fold-specific patterns, validating reproducible learning of ASD-related dysconnectivity.

TABLE IV
DEMOGRAPHIC DETAILS OF STUDIED AGE COHORTS COMPRISING A TOTAL OF 884 SUBJECTS FROM THE ABIDE-1 DATASETS

Age Group	Age Set (Years)	Gender (F/M)	Category (TC/ASD)	Age (Years)
Children	0–10	25/86	62/49	08.81 ± 0.84
Adolescents	10–18	81/410	268/223	13.78 ± 2.13
Adults	18–65	32/250	146/136	26.48 ± 8.03

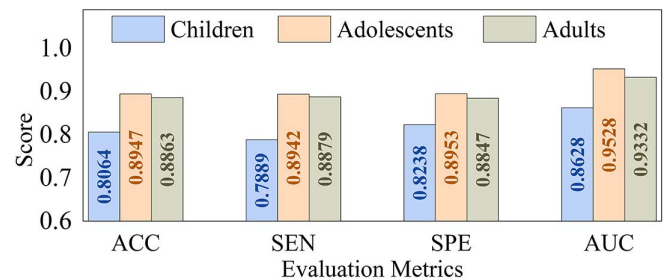


Fig. 8. Illustration of tenfold CV performance comparison between three exclusive age groups for the ABIDE-1 datasets.

The adolescent cohort outperformed the children and adult cohorts in all evaluation metrics, i.e., ACC, SEN, SPE, and AUC with values of 0.8947, 0.8942, 0.8953, and 0.9528, respectively. The second-best results were observed for the adult cohort, and the children cohort exhibited the least impressive performance. Previous studies show mixed findings on age cohorts [27], [34]. In our data (children: 13%, adolescents: 56%, adults: 32%), insufficient children samples limited learning, while the larger adolescent cohort enabled better capture of inter-sample patterns.

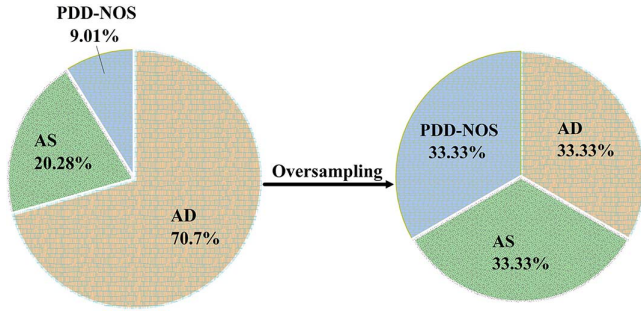


Fig. 9. Oversampling balances severe class imbalance (original ratio 7.84:2.25:1 to balanced 33.33:33.33:33.33) by replicating minority classes (AS, PDD-NOS) to match majority class (AD).

H. Identification of Possible Subtypes

The ASD syndrome can be subcategorized based on DSM_IV_TR criteria, i.e., AD, AS, and PDD-NOS, and so on, as reported in Table I. The identification of these subtypes is an even more challenging task due to the subjectivity involved in clinical presentations and symptom-based diagnosis, which can delay the adoption of specific therapeutic procedures. Inspired by Huang et al. [5], we also leverage the proposed DAMILF as a multiclass classification problem. In this experiment, we utilized the softmax activation function in the output layer instead of sigmoid. The use of the softmax function is intuitively more appropriate in multiclass classification settings because it normalizes the predicted scores/probabilities, and their overall summation is always equal to 1. We also exploited categorical cross-entropy instead of binary cross entropy as a loss function because it is not a binary classification anymore, while the StratifiedKFold module from Scikit-learn library [32] was employed to randomly assign data samples during the tenfold CV. To reduce the bias corresponding to the tenfold CV data split, we also ran this experiment ten times and documented the average values of the evaluation metrics.

Class imbalance (251 AD, 72 AS, 32 PDD-NOS; ratio 7.84: 2.25: 1) required balancing. Data augmentation (rotation, scaling, noise [35]) can corrupt DLL selection and neurological patterns. We used oversampling via scikit-learn resample [32] instead.

The oversampling creates multiple copies of the samples that belong to the minority class. The percentages of all three studied subtypes, before and after the oversampling, are shown in Fig. 9. Moreover, the resampling was performed after the StratifiedKFold module to maintain the class balance and avoid data leakage.

The per-class performance metrics reveal distinct patterns of subtype discriminability as reported in Table V. The AD achieved the highest ACC (0.9216) and F1-measure (0.8846), likely attributable to both the larger original sample size ($n = 251$, requiring no oversampling) and more distinct diagnostic profiles. The AS demonstrated the highest SEN (0.9354) with intermediate ACC (0.9000, $F1 = 0.8617$), indicating that the model readily identifies this subtype. The PDD-NOS exhibited the lowest ACC (0.8898) and notably lower SEN (0.7309), combined with the highest SPE (0.9692). This pattern, where

TABLE V
TENFOLD CV PERFORMANCE FOR DIAGNOSIS OF ASD
SUBTYPES BASED ON THE ABIDE-1 DATASETS

Class	ACC	SEN	SPE	F1-Measure
AD	0.9216	0.9008	0.9320	0.8846
AS	0.9000	0.9354	0.8822	0.8617
PDD-NOS	0.8898	0.7309	0.9692	0.8155

TABLE VI
TENFOLD CV PERFORMANCE COMPARISON BETWEEN OUR DAMILF
MODEL TRAINED WITH KW AND FRIEDMAN CHI-SQUARED MAPS
FOR ASD CLASSIFICATION BASED ON THE ABIDE-1 DATASETS

Method	ACC	SEN	SPE	AUC
KW-based DAMILF	0.8288	0.8286	0.8291	0.8890
FCSM-based DAMILF	0.8556	0.8553	0.8560	0.9201

Note: Bold indicates the best-performing metric values between the two compared configurations.

PDD-NOS is difficult to correctly identify (low SEN) but rarely misclassified when the actual label is another subtype (high SPE), reflects the inherent diagnostic ambiguity of the “not otherwise specified” category. The confusion matrix (Supplementary Table S-VI) reveals that PDD-NOS samples are frequently misclassified as AS (365 cases) or AD (243 cases), demonstrating the overlapping symptomatology. We achieved a macro-average per-class accuracy, macro F1-scores, and balanced accuracy of 0.9038, 0.8539, and 0.8557, respectively. To the best of our knowledge, it is 6.58% higher than the previously best-reported result by Huang et al. [5].

I. Ablation Study

Three ablation studies were conducted to evaluate the effectiveness of the proposed DAMILF model.

1) *FCSMs*: To validate the efficacy of the FCSMs [26] in our model, we did not calculate the voxelwise Friedman chi-squared scores for the KW image. Instead, we directly trained our proposed framework with the KW maps. We performed a tenfold CV comparison of our proposed DAMILF for the ASD diagnosis task with KW images and FCSM maps. The comparative results are presented in Table VI, which demonstrate the improved performance of FCSMs, where the performance gains in terms of ACC, SEN, SPE, and AUC are 2.68%, 2.67%, 2.69%, and 3.11%, respectively.

2) *DMS*: In our proposed model, we removed the DMS that is used to track data-driven reproducible biomarkers through the adopted preprocessing pipeline. Then, we used all 64 patches from a cropped fMRI (i.e., consists of $60 \times 60 \times 60$ voxels) to create a bag. Further, these bags were utilized in model training. A tenfold CV performance comparison of DAMILF for the ASD diagnosis task with and without the DMS is reported in Table VII. The results show that using the DMS achieves performance gains of 21.61%, 22.30%, 22.36%, and 25.26% for the ACC, SEN, SPE, and AUC metrics, respectively. In this experiment, the model was trained with eight patches per bag. We also studied the effect of varying the number of patches per bag, as detailed in Section S-V of the supplementary materials.

TABLE VII
RESULTS OF ASD CLASSIFICATION PRODUCED BY OUR DAMILF METHOD WITH AND WITHOUT THE DMS BY COMPUTING TENFOLD CV PERFORMANCE FOR THE ABIDE-1 DATASETS

Method	ACC	SEN	SPE	AUC
Without DMS	0.6395	0.6323	0.6324	0.6675
With DMS	0.8556	0.8553	0.8560	0.9201

Note: Bold indicates the best-performing metric values between the two compared settings.

It is important to note that the “with DMS” condition used fold-specific templates, recomputed from training data only. The substantial performance degradation without DMS (−0.2161 ACC, −0.2526 AUC) demonstrates that DMS is a critical component of the framework. Training on all 64 patches from uncropped images includes regions with minimal discriminative power, introducing noise and heterogeneity that overwhelm the attention mechanism. By contrast, DMS effectively isolates the eight most informative patches containing statistically significant FC differences (landmark density >50), enabling the dual attention mechanism to focus computational resources on diagnostically relevant patterns. This validates that DMS does not simply reduce computational cost but fundamentally improves feature quality by filtering uninformative brain regions.

3) *Attention Mechanism*: To evaluate the independent and combined contributions of the dual attention mechanisms in DAMILF, we conducted a componentwise ablation study comparing four model variants (Table VIII). The no attention mechanism MIL framework (NA-MILF) baseline removes both attention mechanisms. It achieves ACC = 0.8195, AUC = 0.8663. The substantial gap from DAMILF (+0.0361 ACC, +0.0538 AUC) confirms that attention mechanisms are critical. The feature generation attention mechanism only MIL framework (FGA-MILF) variant retains the spatial attention module within patches (9)–(10), Section IV-C that identifies discriminative FC locations but removes the AWFM adaptive patch weighting. It gets ACC = 0.8246, AUC = 0.8736. The modest improvement over NA-MILF (+0.0051 ACC) indicates that intra-patch spatial attention alone provides limited benefit without inter-patch balancing. The AWFM attention mechanism only MIL framework (AWFM-MILF) variant retains the AWFM (15)–(16), Section IV-D, that adaptively weights patch contributions but removes intra-patch spatial attention. Performance: ACC = 0.8329, AUC = 0.8862. This represents the largest single-component contribution (+0.0134 ACC over NA-MILF), demonstrating that adaptive patch weighting is more impactful than spatial attention alone. Combining both mechanisms, DAMILF yields ACC = 0.8556, AUC = 0.9201, outperforming all single-component variants. The synergistic gain (+0.0227 ACC beyond AWFM-only) demonstrates that spatial and patch-level attention address complementary aspects of feature discrimination: spatial attention refines intra-patch features by emphasizing diagnostically relevant connectivity patterns, while AWFM balances inter-patch contributions by prioritizing information-rich patches. Together, they enable the model to learn multiscale hierarchical features (local within

TABLE VIII
10-FOLD CV RESULTS FOR ASD CLASSIFICATION ACHIEVED BY OUR DAMILF METHOD AND ITS REPLICAS (I.E., WITHOUT ATTENTION MODULES) BASED ON THE ABIDE-1 DATASETS

Method	ACC	SEN	SPE	AUC
NA-MILF	0.8195	0.8183	0.8203	0.8663
FGA-MILF	0.8246	0.8238	0.8248	0.8736
AWFM-MILF	0.8329	0.8319	0.8335	0.8862
DAMILF	0.8556	0.8553	0.8560	0.9201

Note: NA-MILF serves as matched baseline with identical 3-D patch input (8 patches, $15 \times 15 \times 15$), backbone CNN (4 layers, channels 16, 32, 64, and 128), and parameter count (0.56M) as DAMILF, but without dual attention mechanisms. This isolates attention contributions (+0.0361 ACC, +0.0538 AUC).

patches and global across patches) that maximize ASD-TC separability. These results validate that both attention components contribute essential and complementary discriminative power to the DAMILF framework.

VII. DISCUSSION

This section compares DAMILF with previous works, discusses limitations, and suggests future directions. Hyperparameter selection and neighborhood radius (R) analysis are detailed in Supplementary Sections S-III and S-IV, respectively.

A. Performance Evaluation and Methodological Insights

Broader literature comparison (Table S-IX) demonstrates DAMILF’s superior performance against diverse ABIDE-1 methods including graph neural networks, multimodal approaches, and conventional machine learning with varied sample sizes and modalities. The data-driven preprocessing pipeline reduces heterogeneities by filtering less prominent regions, facilitating uniform data representation. Direct comparison baselines (Table II) enable controlled evaluation (Supplementary Section S-IX). Performance improvements (+0.0435 ACC, +0.0664 AUC) justify architectural complexity through interpretability, biological motivation, and computational efficiency.

The neuroanatomical findings in Section VI-F provide evidence for DAMILF’s interpretability through convergence with established ASD neurobiology. The autonomously identified attention-weighted regions correspond closely to brain networks showing characteristic functional dysconnectivity in autism literature [8], [24]. Temporal cortex involvement (37.5% of attention) aligns with documented deficits in social cognition and language processing [24], while insular dysfunction (25.0%) relates to impaired interoception and emotional regulation in ASD populations. The observed right hemisphere predominance (12–20 regions) is consistent with lateralization theories of atypical connectivity in autism. The high cross-fold stability (mean CoV = 12.15%) and strong correlation with landmark density ($r = 0.997$) demonstrate that attention patterns are reproducible biomarkers rather than fold-specific artifacts. Notably, patch 35 (insula) exhibited highest per-landmark efficiency despite moderate landmark density, suggesting that

the dual attention mechanism captures context-dependent feature importance beyond simple voxel counting. Unlike atlas-dependent ROI approaches [10], [13], [20], our data-driven landmarks autonomously converge on clinically meaningful regions, indicating that ASD-related dysconnectivity patterns intrinsically guide the attention allocation without requiring anatomical priors. The ablation study in Table VII confirms the critical contribution of DMS (+0.2161 ACC), validating that data-driven landmark identification effectively isolates discriminative connectivity patterns.

The clinical utility of DAMILF extends beyond performance metrics. Current ASD diagnosis relies predominantly on behavioral assessments (DSM criteria) [4], which often result in diagnostic delays of 2–4 years post symptom onset, particularly in lower resource settings or for individuals without overt behavioral presentations [3]. Our framework achieves 0.8556 ACC (AUC: 0.9201) with balanced SEN (0.8553) and SPE (0.8560) using only routine 5–10 min rs-fMRI scans, meeting suggested thresholds for objective screening tools (>80% accuracy with balanced error rates) [36]. Importantly, DAMILF could serve as a complementary screening instrument to prioritize cases for comprehensive clinical evaluation, potentially reducing diagnostic delays and enabling earlier intervention access. The balanced false negative (14.47%) and false positive (14.40%) rates indicate that neither error type is disproportionately favored, which is appropriate for screening contexts where follow-up clinical assessment remains the gold standard. We emphasize that DAMILF should augment, not replace, expert clinical judgment.

DAMILF differs from previous frameworks [5], [10], [12], [13], [20], [37], [38], [39], [40], [41], [42]: while prior methods depend on preidentified ROIs, we use DMS for region selection. Our method extracts patch-based local information via patch-level attention, refined by AWFMs with adaptive weighting, while SSNs capture global features, enabling multiscale feature exploitation.

Beyond these method-specific comparisons, our investigation yields generalizable insights for attention-based MIL in neuroimaging: hierarchical multiscale attention provides synergistic gains exceeding individual components, data-driven region selection substantially outperforms atlas-based approaches, and multisite robustness requires explicit heterogeneity-handling strategies beyond attention mechanisms alone. These design principles demonstrated through our ablation studies (Table VIII), multisite validation (Table III), and interpretability analysis (Section VI-F); generalize to other neuroimaging applications facing spatial heterogeneity and multisite protocols.

B. Limitations and Future Work

We acknowledge that our accuracy improvement appears modest (+0.0435); however, our AUC of 0.9201 for a single model framework is the highest reported for ABIDE-1 whole-dataset classification. The combination of performance gains, interpretability, and generalizability justifies the architectural

complexity (refer to Supplementary Section S-IX for detailed analysis).

A primary limitation is the absence of external validation beyond the ABIDE-1 consortium. While LOSOCV (Table III) demonstrates generalization across 17 independent sites with heterogeneous scanner types, acquisition protocols, and subject demographics, validation on completely independent datasets collected outside the ABIDE initiative is essential for clinical translation. External validation would test generalization to different populations, recruitment strategies, and diagnostic assessment procedures beyond those represented in ABIDE-1. Most comparable studies reported in Table S-IX similarly rely on single-consortium validation. Future work will prioritize external validation to demonstrate reproducibility across diverse clinical settings and independently collected datasets.

First, the input patch size is fixed for all DLLs, but the neurological disruptions triggered by ASD may appear at multiple scales and in more than one region. Consequently, a better solution resides in developing multiscale and variable size input patches, but it may aggravate problems associated with model configuration. Moreover, multiscale patch-based approaches could be a possible fix to select flexible patch sizes [43]. Additionally, while our DMS is empirically validated through patch number sensitivity analysis (4, 8, 12, 16, 32 patches in Supplementary Fig. S-4), future work should explore: 1) subject-adaptive patch selection strategies where the number of patches varies per subject based on individual landmark distributions, rather than using a fixed patch configuration for all subjects; and 2) alternative patch selection methodologies beyond density-based ranking. Such analyses would provide additional evidence regarding the broader applicability of the data-driven landmark approach across diverse neuroimaging protocols, preprocessing choices, and selection strategies.

Second, the multisite ABIDE-1 datasets utilize independent scanners and data acquisition protocols, which result in diverse data distributions. Thus, embedding a domain adaptation strategy into the present framework could further enhance its generalizability [44].

Third, we used the oversampling of minority class to balance class distribution for ASD subtype identification. In our future work, the data augmentation module can be employed in addition to oversampling to reduce bias in the datasets [28]. Misclassification carries distinct patient welfare implications: false negatives risk delayed intervention during critical developmental windows, while false positives may cause parental anxiety. Responsible clinical integration requires clinician training protocols, clear accountability frameworks when artificial intelligence (AI) recommendations diverge from clinical judgment, and explicit validation across socioeconomic and demographic populations to ensure equitable access and prevent exacerbating existing healthcare disparities in ASD diagnosis. Furthermore, possibly a single modality will not be sufficient to represent all nonlinear spatiotemporal heterogeneities of multisite 4-D fMRIs. Hence, the current method could be modified to construct a multimodal framework [27], which could further improve the diagnosis performance.

VIII. CONCLUSION

This article presents DAMILF, a dual attention multi-instance learning framework for automated ASD diagnosis from resting-state fMRI. Evaluated on 884 ABIDE-1 subjects across 17 sites, the method achieved 0.8556 accuracy (AUC: 0.9201) with robust cross-site generalization. Ablation studies quantified component contributions: data-driven masking improved accuracy by 0.2161, dual attention by 0.0361, and FCSM by 0.0268. Learned attention patterns converged on established ASD-affected brain regions, demonstrating biological interpretability. The framework provides an objective tool to complement clinical ASD assessment.

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